DSA0603 - DATA HANDLING AND VISUALIZATION

ACTIVITY-2

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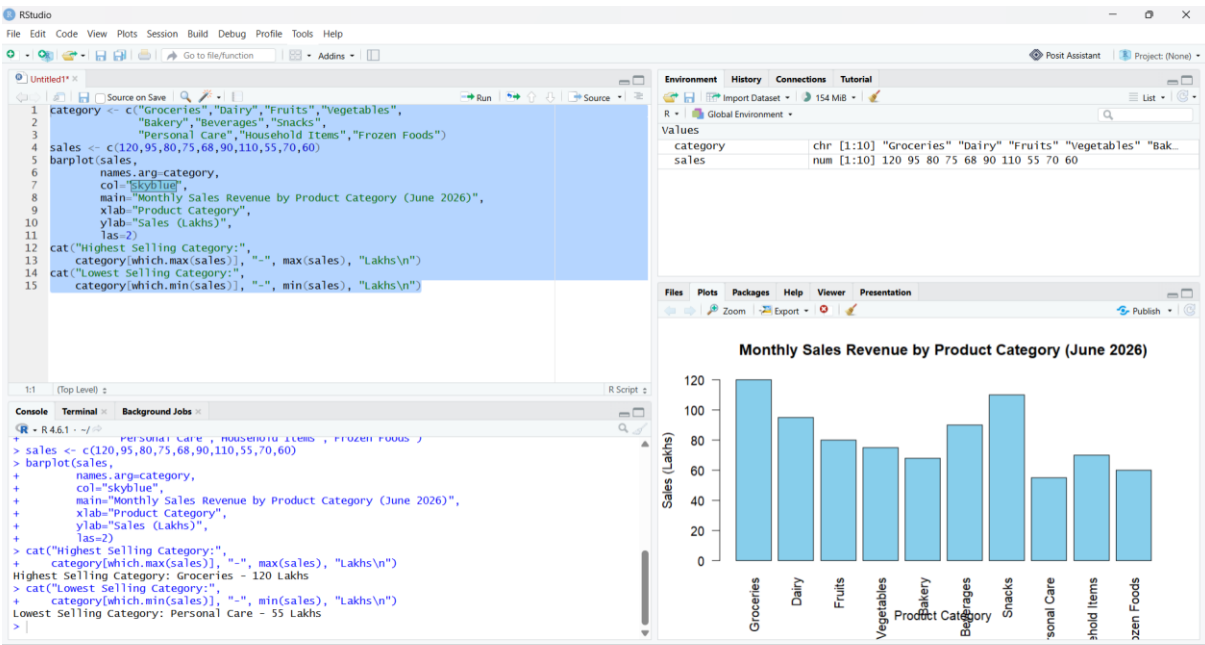
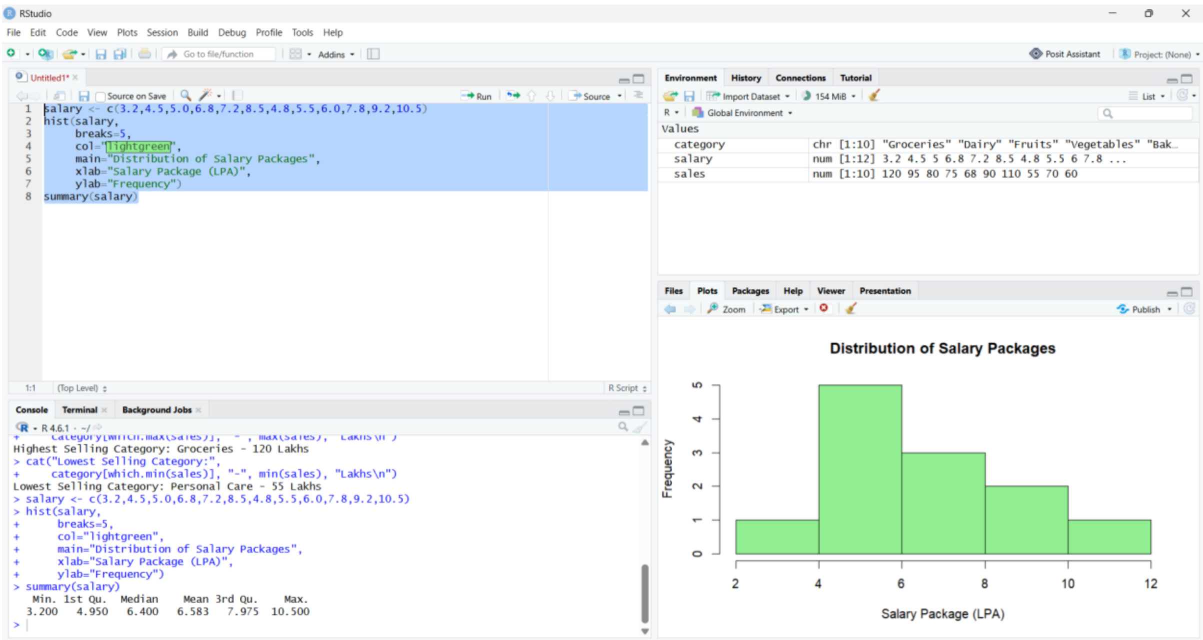
Q1. Bar Chart of Product

```
category <- c("Groceries","Dairy","Fruits","Vegetables",
           "Bakery","Beverages","Snacks",
           "Personal Care","Household Items","Frozen Foods")

sales <- c(120,95,80,75,68,90,110,55,70,60)

barplot(sales,
      names.arg=category,
      col="skyblue",
      main="Monthly Sales Revenue by Product Category (June 2026)",
      xlab="Product Category",
      ylab="Sales (Lakhs)",
      las=2)

cat("Highest Selling Category:",
   category[which.max(sales)], "-", max(sales), "Lakhs\n")

cat("Lowest Selling Category:",
   category[which.min(sales)], "-", min(sales), "Lakhs\n")
```



Q2.Histogram of Salary

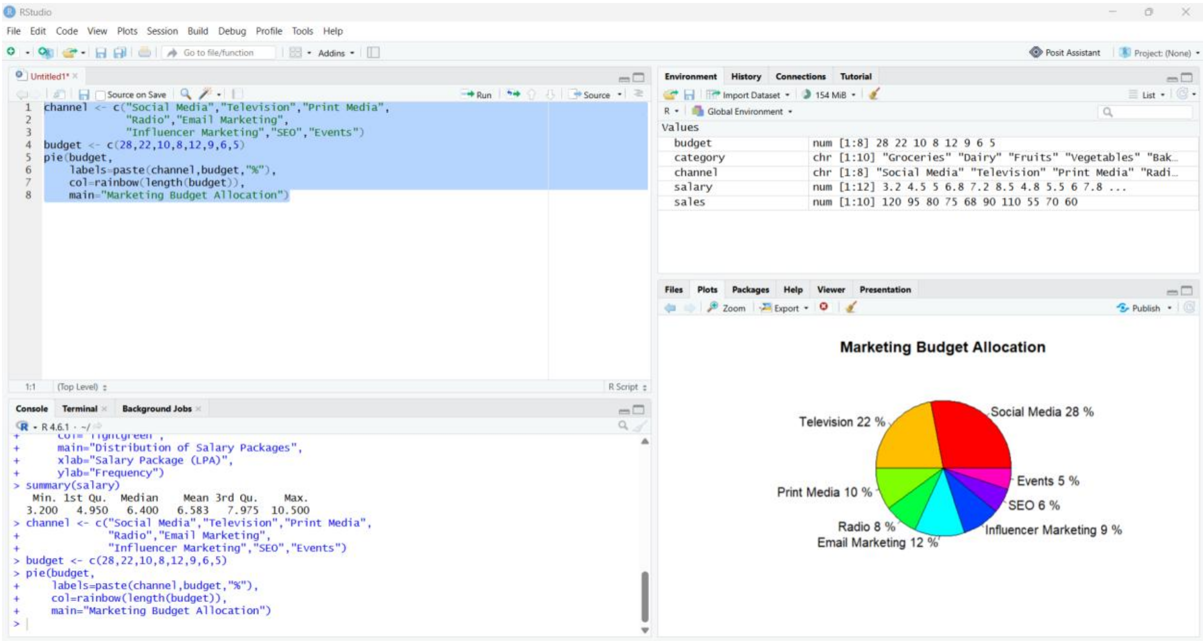
```
salary <- c(3.2,4.5,5.0,6.8,7.2,8.5,4.8,5.5,6.0,7.8,9.2,10.5)
hist(salary,
    breaks=5,
    col="lightgreen",
    main="Distribution of Salary Packages",
    xlab="Salary Package (LPA)",
    ylab="Frequency")
summary(salary)
```



Q3. Pie Chart of Marketing Channels

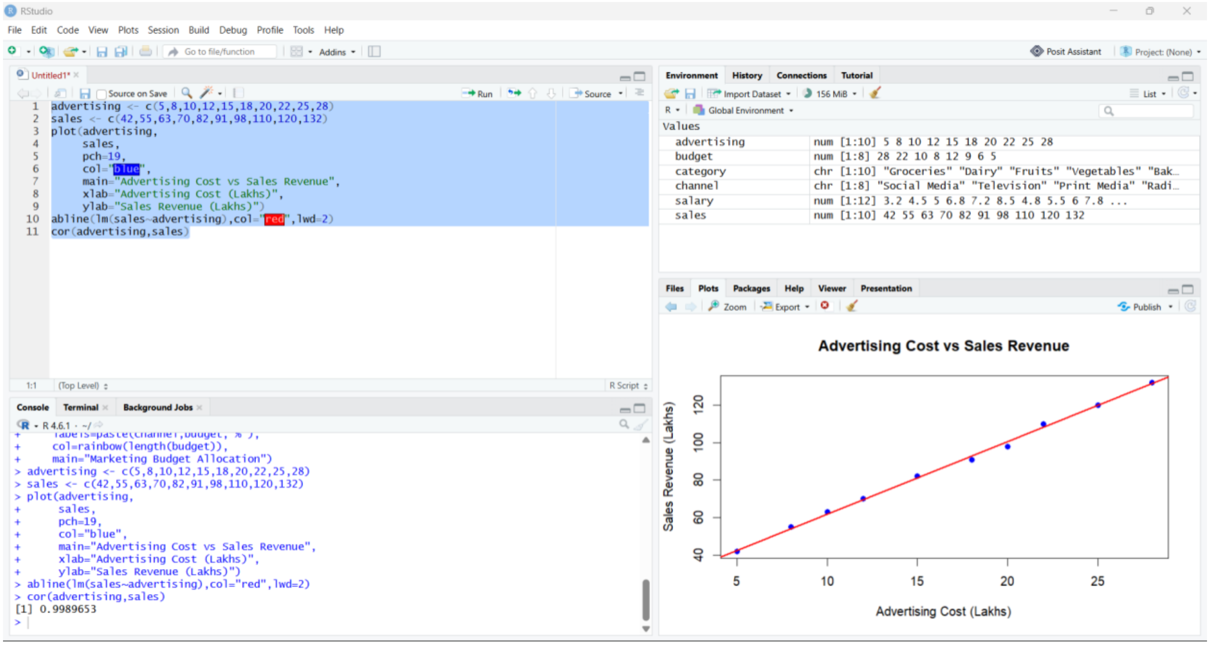
```
channel <- c("Social Media","Television","Print Media",
        "Radio","Email Marketing",
        "Influencer Marketing","SEO","Events")
```

```
budget <- c(28,22,10,8,12,9,6,5)

pie(budget,
   labels=paste(channel,budget,"%"),
   col=rainbow(length(budget)),
   main="Marketing Budget Allocation")
```



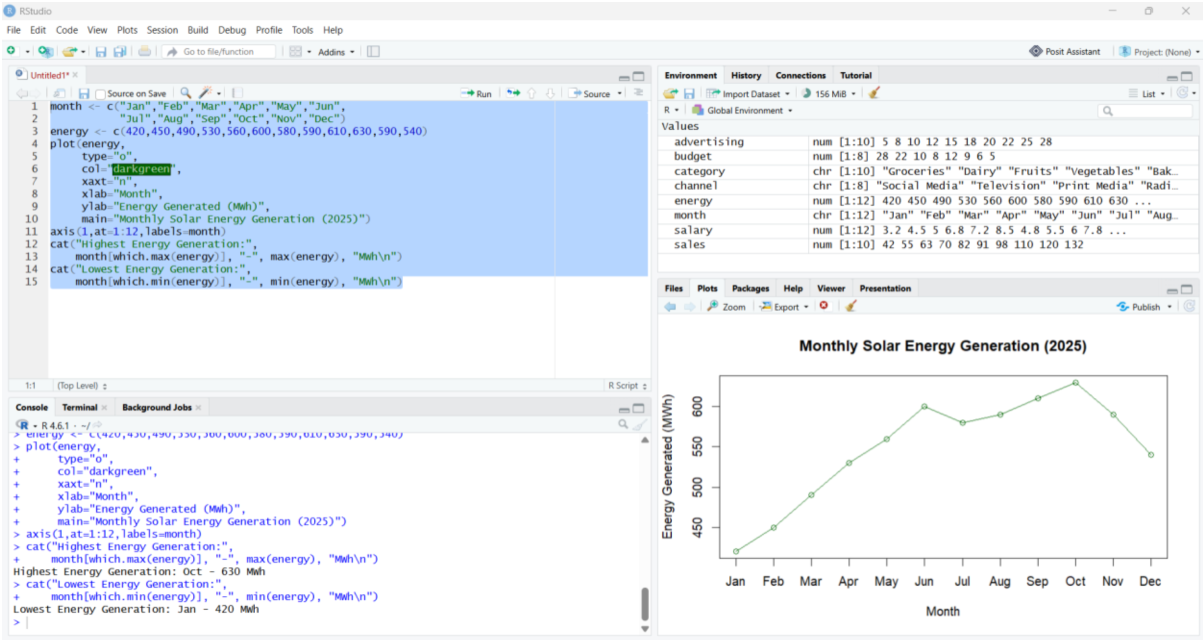
Q4.Scatter Plot of Advertising

```
advertising <- c(5,8,10,12,15,18,20,22,25,28)

sales <- c(42,55,63,70,82,91,98,110,120,132)

plot(advertising,
    sales,
    pch=19,
    col="blue",
    main="Advertising Cost vs Sales Revenue",
    xlab="Advertising Cost (Lakhs)",
    ylab="Sales Revenue (Lakhs)")

abline(lm(sales~advertising),col="red",lwd=2)

cor(advertising,sales)
```



Q5.Line Chart of Energy Generation

```
month <- c("Jan","Feb","Mar","Apr","May","Jun",
         "Jul","Aug","Sep","Oct","Nov","Dec")
energy <- c(420,450,490,530,560,600,580,590,610,630,590,540)
plot(energy,
    type="o",
    col="darkgreen",
    xaxt="n",
    xlab="Month",
    ylab="Energy Generated (MWh)",
    main="Monthly Solar Energy Generation (2025)")
axis(1,at=1:12,labels=month)
cat("Highest Energy Generation:",
   month[which.max(energy)], "-", max(energy), "MWh\n")
cat("Lowest Energy Generation:",
   month[which.min(energy)], "-", min(energy), "MWh\n")
```


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
Untitled1*
Source on Save
Run
Source
month <- c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec")
energy <- c(420,450,490,530,560,600,580,590,610,630,590,540)
plot(energy,
type="o",
col="darkgreen",
xaxt="n",
xlab="Month",
ylab="Energy Generated (MWh)",
main="Monthly Solar Energy Generation (2025)")
axis(1,at=1:12,labels=month)
cat("Highest Energy Generation:",
month[which.max(energy)], "-", max(energy), "MWh\n")
cat("Lowest Energy Generation:",
month[which.min(energy)], "-", min(energy), "MWh\n")
1:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> plot(energy,
+ type="o",
+ col="darkgreen",
+ xaxt="n",
+ xlab="Month",
+ ylab="Energy Generated (MWh)",
+ main="Monthly Solar Energy Generation (2025)")
> axis(1,at=1:12,labels=month)
> cat("Highest Energy Generation:",
+ month[which.max(energy)], "-", max(energy), "MWh\n")
Highest Energy Generation: Oct - 630 MWh
> cat("Lowest Energy Generation:",
+ month[which.min(energy)], "-", min(energy), "MWh\n")
Lowest Energy Generation: Jan - 420 MWh
Environment
History
Connections
Tutorial
Import Dataset
156 MiB
List
Global Environment
Values
advertising num [1:10] 5 8 10 12 15 18 20 22 25 28
budget num [1:8] 28 22 10 8 12 9 6 5
category chr [1:10] "Groceries" "Dairy" "Fruits" "Vegetables" "Bak…
channel chr [1:8] "Social Media" "Television" "Print Media" "Radi…
energy num [1:12] 420 450 490 530 560 600 580 590 610 630 ...
month chr [1:12] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug…
salary num [1:12] 3.2 4.5 5 6.8 7.2 8.5 4.8 5.5 6 7.8 ...
sales num [1:10] 42 55 63 70 82 91 98 110 120 132
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Monthly Solar Energy Generation (2025)
Energy Generated (MWh)
450
500
550
600
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Month